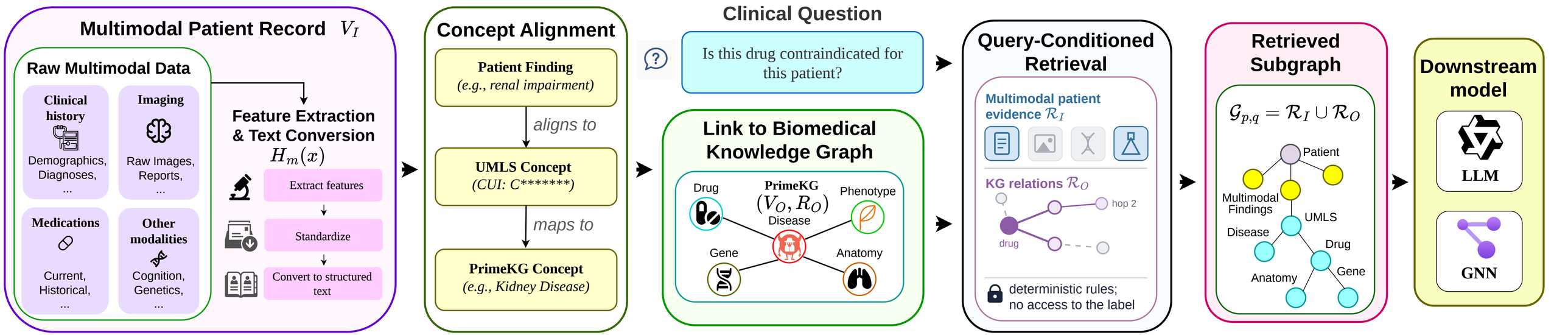

Multimodal Patient Record V_I
Raw Multimodal Data
Clinical history
Demographics, Diagnoses, ...
Imaging
Raw Images, Reports, ...
Medications
Current, Historical, ...
Other modalities
Cognition, Genetics, ...
Feature Extraction & Text Conversion
$H_m(x)$
Extract features
Standardize
Convert to structured text
Concept Alignment
Patient Finding
(e.g., renal impairment)
aligns to
UMLS Concept
(CUI: C*******)
maps to
PrimeKG Concept
(e.g., Kidney Disease)
Clinical Question
Is this drug contraindicated for this patient?
Link to Biomedical Knowledge Graph
Drug
PrimeKG
(V_O, R_O)
Phenotype
Disease
Gene
Anatomy
Query-Conditioned Retrieval
Multimodal patient evidence $\mathcal{R}_I$
KG relations $\mathcal{R}_O$
hop 2
drug
deterministic rules; no access to the label
Retrieved Subgraph
$\mathcal{G}_{p,q} = \mathcal{R}_I \cup \mathcal{R}_O$
Patient
Multimodal Findings
UMLS
Disease
Drug
Anatomy
Gene
Downstream model
LLM
GNN